

Algorithmic Documentation - Zoning Plan Generator

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1. Algorithm Overview

1.1 High-Level Architecture

ZONING PLAN GENERATOR

Phase 1: Data Generation

- Generate synthetic blocks
- Create superblock clusters
- Calculate target compositions

Phase 2: Grid Initialization

- Create W×H grid
- Place building features (cores)
- Mark restricted areas

Phase 3: Block Placement

- Sort blocks by priority
- Iterate through superblocks
- Place blocks with adjacency check
- Apply greedy heuristic

- Phase 4: Constraint Analysis
- Calculate geometric scores
 - Evaluate adjacency compliance
 - Measure composition matches
 - Detect violations

- Phase 5: Output Generation
- Export to JSON/CSV
 - Generate visualizations
 - Create analysis reports

1.2 Algorithm Classification

- **Type:** Greedy heuristic with constraint satisfaction
 - **Paradigm:** Constructive placement with backtracking
 - **Optimization:** Multi-objective with priority hierarchy
 - **Determinism:** Pseudo-random (with seed control)
 - **Completeness:** Incomplete (may not find optimal solution)
 - **Soundness:** Sound (all solutions satisfy hard constraints)
-

2. Data Structures

2.1 Block Structure

```
class Block:
    id: String           # Unique identifier "BLK-XXX"
    name: String         # Block type name
    category: Enum       # {ME, WE, US, SUPPORT, SPECIALTY}
    width: Integer       # Width in meters
    height: Integer      # Height in meters
    area: Float          # Computed area (width × height)
    organization: String # "Anchor Resident"
    superblock_id: String # Assigned superblock "SB-XX"
    placed: Boolean      # Placement status
    x: Integer | Null    # X-coordinate on grid
    y: Integer | Null    # Y-coordinate on grid
```

Memory: ~200 bytes per block

Indexing: HashMap by ID for O(1) lookup

Ordering: Array sorted by category priority

2.2 Superblock Structure

```
class Superblock:
    id: String                    # "SB-XX"
    blocks: Array<Block>          # Member blocks
    target_composition: Dict<Float> # Target ratios per category
    actual_composition: Dict<Float> # Actual ratios
    composition_match: Float       # Match score [0, 100]

    # Computed properties
    size: Integer                 # Number of blocks
    total_area: Float             # Sum of block areas
    centroid: Tuple<Float>        # Center position
```

Memory: ~500 bytes + references

Indexing: HashMap by ID

Relationships: One-to-many with blocks

2.3 Grid Structure

```
class Grid:
    width: Integer                # Grid width (default: 100)
    height: Integer               # Grid height (default: 80)
    cells: Array[height][width] # 2D array of cell IDs

    # Each cell contains:
    # - null (empty)
    # - Block ID
    # - Feature ID
```

Memory: $W \times H \times 8$ bytes (pointer size)

For 100×80: ~64 KB

Access: O(1) random access

Iteration: O($W \times H$)

2.4 Adjacency Rule Structure

```
class AdjacencyRule:
    priority: Integer             # [0-5]
    from_block: String            # Source block type
    to_block: String              # Target block/feature
    relation: Enum                # {must_avoid, must_be_near,
    # preferred_near, should_avoid}
    weight: Float                 # Score modifier [-2, 1]
```

Storage: Array sorted by priority
Lookup: Linear scan (small dataset ~20 rules)
Application: Lazy evaluation during placement

2.5 Building Feature Structure

```
class BuildingFeature:
    id: String          # "CORE-A", "MAIN-ENTRY"
    x: Integer          # X position
    y: Integer          # Y position
    width: Integer      # Feature width
    height: Integer     # Feature height
    type: Enum          # {core, anchor, special}
```

Storage: Array (fixed size ~10-15 features)
Spatial Index: R-tree for range queries (optional)

3. Core Algorithms

3.1 Synthetic Block Generation

Algorithm: GENERATE_BLOCKS

INPUT:

- count: Number of blocks to generate
- block_types: Dictionary of categories to block type lists
- dimension_ranges: Dimension constraints by category

OUTPUT:

- blocks: Array of generated Block objects

ALGORITHM:

1. Initialize empty blocks array
2. FOR i = 1 TO count:
 - a. Select random category from available categories
 - b. Select random block type from category
 - c. Get dimension range for category
 - d. Generate width, height based on range
 - e. Apply rectangularity preference (70% similar dimensions)
 - f. Calculate area = width × height
 - g. Create Block object with ID = "BLK-" + padded(i)
 - h. Add block to array
3. RETURN blocks

TIME: $O(n)$

SPACE: $O(n)$

Dimension Generation Detail:

```
FUNCTION generate_dimensions(category):
    range = DIMENSION_RANGES[category]
    (w_min, w_max, h_min, h_max) = range

    IF random() < 0.7: // 70% chance of rectangular
        width = random_int(w_min, w_max)
        IF random() < 0.5:
            height = width
        ELSE:
            height = random_int(h_min, h_max)
    ELSE:
        width = random_int(w_min, w_max)
        height = random_int(h_min, h_max)

    RETURN (width, height)
```

3.2 Superblock Formation

Algorithm: CREATE_SUPERBLOCKS

INPUT:

- blocks: Array of all blocks
- compositions: Array of target composition templates
- min_size: Minimum superblock size (default: 20)
- max_size: Maximum superblock size (default: 30)

OUTPUT:

- superblocks: Array of Superblock objects

ALGORITHM:

1. Initialize empty superblocks array
2. unassigned = copy(blocks)
3. superblock_id = 1
4. WHILE length(unassigned) > min_size AND superblock_id ≤ 12:
 - a. target_size = random_int(min_size, max_size)
 - b. target_comp = random_choice(compositions)
 - c. selected_blocks = []
 - d. FOR each category in [ME, WE, US, SUPPORT, SPECIALTY]:
 - i. needed = floor(target_size × target_comp[category])
 - ii. available = filter(unassigned, category)
 - iii. take = min(needed, length(available))
 - iv. selection = first(available, take)
 - v. Assign superblock_id to each in selection

```

vi. Append selection to selected_blocks
vii. Remove selection from unassigned

e. IF length(selected_blocks) > 0:
i. Calculate actual_composition
ii. Calculate match_score
iii. Create Superblock object
iv. Add to superblocks array
v. superblock_id++

5. RETURN superblocks

TIME:  $O(n \times m)$  where  $m$  = number of superblocks
SPACE:  $O(n + m)$ 

Composition Matching:

FUNCTION calculate_composition_match(target, actual):
categories = [ME, WE, US, SUPPORT, SPECIALTY]
total_error = 0

FOR category IN categories:
target_val = target[category]
actual_val = actual[category]
error = abs(target_val - actual_val)
total_error += error

match = max(0, 1 - total_error) × 100
RETURN match

```

3.3 Grid Initialization

Algorithm: INITIALIZE_GRID

```

INPUT:
- width: Grid width
- height: Grid height
- features: Array of BuildingFeature objects

OUTPUT:
- grid: 2D array initialized with features

ALGORITHM:
1. Create 2D array grid[height][width] filled with null

2. FOR each feature IN features:
a. (x, y) = feature position

```

```

    b. (w, h) = feature dimensions

    c. FOR dy = 0 TO h-1:
        FOR dx = 0 TO w-1:
            grid[y + dy][x + dx] = feature.id

3. RETURN grid

TIME:  $O(W \times H + \Sigma(\text{feature\_area}))$ 
SPACE:  $O(W \times H)$ 

```

3.4 Main Placement Algorithm

Algorithm: PLACE_BLOCKS

INPUT:

- blocks: Array of all blocks
- superblocks: Array of superblocks
- grid: Initialized grid
- adjacency_rules: Array of rules

OUTPUT:

- placed_blocks: Array of successfully placed blocks
- updated_grid: Grid with placed blocks

ALGORITHM:

1. placed_blocks = []
2. Sort superblocks by ID
3. FOR each superblock IN superblocks:
 - a. Determine starting position (superblock_x, superblock_y)
 - b. Sort blocks in superblock by category priority
 - c. current_x = superblock_x
 - d. current_y = superblock_y
 - e. max_row_height = 0
 - f. FOR each block IN superblock.blocks:
 - i. placed = false
 - ii. attempts = 0
 - iii. max_attempts = 100
 - iv. WHILE NOT placed AND attempts < max_attempts:
 1. IF current_x + block.width > grid.width - 5:
 current_x = superblock_x
 current_y += max_row_height + 1
 max_row_height = 0

```

2. IF current_y + block.height > grid.height - 5:
    BREAK // Out of space

3. IF can_place(block, current_x, current_y, grid):
    score = evaluate_adjacency(block, current_x,
                                current_y, rules)

    IF score >= threshold:
        mark_grid(grid, block, current_x, current_y)
        block.x = current_x
        block.y = current_y
        block.placed = true
        placed_blocks.append(block)
        max_row_height = max(max_row_height, block.height)
        current_x += block.width + 1
        placed = true
    ELSE:
        current_x += 2
ELSE:
    current_x += 1

4. attempts++

4. RETURN (placed_blocks, grid)

TIME:  $O(n \times k \times |R| \times |F|)$  where  $k$  = avg attempts
SPACE:  $O(n + W \times H)$ 

```

3.5 Collision Detection

Algorithm: CAN_PLACE

INPUT:

- block: Block to place
- x, y: Candidate position
- grid: Current grid state

OUTPUT:

- valid: Boolean indicating if placement is valid

ALGORITHM:

1. // Check bounds
 - IF $x < 0$ OR $x + \text{block.width} > \text{grid.width}$:
 - RETURN false
 - IF $y < 0$ OR $y + \text{block.height} > \text{grid.height}$:

```

        RETURN false

2. // Check cell availability
   FOR dy = 0 TO block.height - 1:
       FOR dx = 0 TO block.width - 1:
           IF grid[y + dy][x + dx] IS NOT null:
               RETURN false

3. RETURN true

TIME:  $O(\text{block.width} \times \text{block.height})$ 
SPACE:  $O(1)$ 

```

3.6 Adjacency Evaluation

Algorithm: EVALUATE_ADJACENCY

INPUT:

- block: Block to evaluate
- x, y: Candidate position
- rules: Array of adjacency rules
- features: Array of building features

OUTPUT:

- score: Numeric adjacency score

ALGORITHM:

```

1. score = 0
2. position = (x, y)

3. FOR each rule IN rules:
    IF rule.from_block matches block.name:
        // Find target features
        targets = find_targets(rule.to_block, features)

        FOR each target IN targets:
            distance = euclidean_distance(position, target.position)

            // Apply relation-specific scoring
            IF rule.relation == "must_avoid":
                IF distance < 10:
                    score += rule.weight × 10 // Heavy penalty
                ELSE:
                    score += 2 // Reward

            ELSE IF rule.relation == "must_be_near":

```

```

        IF distance < 15:
            score += abs(rule.weight) × 3
        ELSE IF distance > 30:
            score -= 5

    ELSE IF rule.relation == "preferred_near":
        IF distance < 20:
            score += rule.weight × 2
        ELSE IF distance < 30:
            score += rule.weight

    ELSE IF rule.relation == "should_avoid":
        IF distance > 5:
            score += abs(rule.weight)

4. RETURN score

TIME:  $O(|R| \times |F|)$ 
SPACE:  $O(1)$ 

```

3.7 Distance Calculation

Algorithm: EUCLIDEAN_DISTANCE

INPUT:

- p1: First position (x1, y1)
- p2: Second position (x2, y2)

OUTPUT:

- distance: Euclidean distance

ALGORITHM:

1. $dx = x2 - x1$
2. $dy = y2 - y1$
3. $distance = \sqrt{dx^2 + dy^2}$
4. RETURN distance

TIME: $O(1)$

SPACE: $O(1)$

4. Pseudocode

4.1 Complete Main Algorithm

ALGORITHM ZoningPlanGenerator

```

PROCEDURE Main():
    // Phase 1: Data Generation
    config = LoadConfiguration()
    blocks = GenerateBlocks(count=100, config)
    superblocks = CreateSuperblocks(blocks, config.compositions)

    // Phase 2: Grid Initialization
    grid = InitializeGrid(width=100, height=80)
    features = CreateBuildingFeatures()
    PlaceFeatures(grid, features)

    // Phase 3: Block Placement
    adjacency_rules = LoadAdjacencyRules()
    (placed_blocks, updated_grid) = PlaceBlocks(blocks, superblocks,
                                                grid, adjacency_rules)

    // Phase 4: Analysis
    geometric_score = AnalyzeGeometry(placed_blocks)
    adjacency_scores = AnalyzeAdjacency(placed_blocks, adjacency_rules,
                                       features)
    composition_scores = AnalyzeCompositions(superblocks)
    violations = DetectViolations(placed_blocks, adjacency_rules, features)

    // Phase 5: Output
    analysis_report = CreateReport(geometric_score, adjacency_scores,
                                  composition_scores, violations)
    ExportData(blocks, superblocks, features, analysis_report)
    GenerateVisualizations(updated_grid, blocks, superblocks, features)

    RETURN (blocks, superblocks, updated_grid, analysis_report)

END PROCEDURE

```

4.2 Greedy Placement with Backtracking

```

PROCEDURE PlaceBlockGreedy(block, superblock, grid, rules):
    start_x, start_y = GetSuperblockAnchor(superblock)
    best_position = null
    best_score = -infinity

    // Search window
    FOR y = start_y TO min(start_y + 50, grid.height - block.height):
        FOR x = start_x TO min(start_x + 50, grid.width - block.width):

            IF CanPlace(block, x, y, grid):
                score = EvaluateAdjacency(block, x, y, rules)

```

```

        IF score > best_score:
            best_score = score
            best_position = (x, y)

        // Early acceptance for good scores
        IF score >= threshold AND score > 0:
            RETURN (x, y, score)

    // Use best found position if above minimum threshold
    IF best_score >= minimum_threshold:
        RETURN best_position, best_score

    RETURN null, -infinity
END PROCEDURE

```

4.3 Geometric Rectangularity Analysis

```

PROCEDURE AnalyzeGeometry(blocks):
    rectangular_count = 0
    total_placed = 0

    FOR each block IN blocks:
        IF block.placed:
            total_placed++
            aspect_ratio = max(block.width, block.height) /
                           min(block.width, block.height)

            IF aspect_ratio <= 2.0:
                rectangular_count++

    IF total_placed == 0:
        RETURN 0

    score = (rectangular_count / total_placed) × 100
    RETURN score
END PROCEDURE

```

4.4 Adjacency Compliance Analysis

```

PROCEDURE AnalyzeAdjacency(blocks, rules, features):
    compliance = {}

    // Group rules by priority
    FOR priority = 0 TO 5:
        priority_rules = filter(rules, r => r.priority == priority)

```

```

    compliant_count = 0
    total_count = 0

    FOR each rule IN priority_rules:
        applicable_blocks = filter(blocks, b => matches(b, rule.from_block))

        FOR each block IN applicable_blocks:
            IF block.placed:
                total_count++

                targets = FindTargets(rule.to_block, features)
                IF CheckRuleSatisfaction(block, targets, rule):
                    compliant_count++

    IF total_count > 0:
        compliance[priority] = (compliant_count / total_count) * 100
    ELSE:
        compliance[priority] = 100

    RETURN compliance
END PROCEDURE

```

5. Complexity Analysis

5.1 Time Complexity Breakdown

Phase 1: Data Generation

GenerateBlocks(n):	$O(n)$
CreateSuperblocks(n, m):	$O(n \times m)$
Total:	$O(n \times m)$

Phase 2: Grid Initialization

InitializeGrid(W, H):	$O(W \times H)$
PlaceFeatures(F):	$O(\sum \text{feature_area})$
Total:	$O(W \times H)$

Phase 3: Block Placement

For each block:	n iterations
Search positions:	$O(k)$ average attempts
CanPlace check:	$O(w \times h)$ per attempt
EvaluateAdjacency:	$O(R \times F)$
Per block:	$O(k \times (w \times h + R \times F))$
Total placement:	$O(n \times k \times (w \times h + R \times F))$

Best case:	$\Omega(n \times R \times F)$
Average case:	$\Theta(n \times k \times R \times F)$
Worst case:	$O(n \times W \times H \times R \times F)$

Phase 4: Analysis

AnalyzeGeometry:	$O(n)$
AnalyzeAdjacency:	$O(n \times R \times F)$
AnalyzeCompositions:	$O(m \times n)$
Total:	$O(n \times (m + R \times F))$

Phase 5: Output

ExportJSON:	$O(n + m)$
ExportCSV:	$O(n)$
GenerateVisualizations:	$O(W \times H + n)$
Total:	$O(W \times H + n + m)$

Overall Time Complexity:

$$T(n, m, W, H, R, F, k) = O(n \times k \times (w \times h + |R| \times |F|) + W \times H)$$

Typical values:

$$n = 100, m = 12, W = 100, H = 80, |R| = 20, |F| = 15, k = 10$$

$$T = O(100 \times 10 \times (9 + 300) + 8000) = O(300,000) \text{ operations}$$

Estimated runtime: 2-3 seconds

5.2 Space Complexity Breakdown

Data Structures:

Blocks:	$O(n) \times 200 \text{ bytes}$	20 KB
Superblocks:	$O(m) \times 500 \text{ bytes}$	6 KB
Grid:	$O(W \times H) \times 8 \text{ bytes}$	64 KB
Features:	$O(F) \times 100 \text{ bytes}$	1.5 KB
Rules:	$O(R) \times 100 \text{ bytes}$	2 KB
Total:	$O(W \times H + n + m + R + F)$	93.5 KB

Auxiliary Space:

Temporary arrays:	$O(n)$	
Search state:	$O(k)$	
Analysis results:	$O(m + R)$	
Total auxiliary:	$O(n + k + m + R)$	15 KB

Overall Space Complexity:

$$S(n, m, W, H, R, F) = O(W \times H + n + m + |R| + |F|)$$

Typical: 108.5 KB

Peak memory: < 1 MB

5.3 Scalability Analysis

Linear Scaling (blocks):

$n \rightarrow 2n$: Time $\times 2$, Space $\times 1.2$

Quadratic Scaling (grid):

$W, H \rightarrow 2W, 2H$: Time $\times 4$, Space $\times 4$

Practical Limits:

Maximum blocks:	~1000 (10 seconds runtime)
Maximum grid:	200×200 (16 MB memory)
Maximum rules:	~100 (minimal impact)

6. Optimization Strategies

6.1 Spatial Indexing

R-Tree for Features:

STRUCTURE RTree:

- Build R-tree index for building features
- Query: FindFeaturesInRadius(x, y, radius)
- Time: $O(\log|F| + k)$ where k = results

Benefit: Reduce adjacency checks from $O(|F|)$ to $O(\log|F|)$

Impact: 40-60% speedup for large feature sets

6.2 Early Termination

Score Threshold:

IF score > excellent_threshold:

- ACCEPT immediately
- SKIP further search

Threshold values:

- excellent_threshold = 10
- good_threshold = 5
- acceptable_threshold = -1

Benefit: Average 30% reduction in placement time

6.3 Caching

Distance Cache:

CACHE distance_cache[block_id][feature_id]

IF distance already computed:

 RETURN cached_value

ELSE:

 distance = compute()

 STORE in cache

 RETURN distance

Benefit: Eliminate repeated distance calculations

6.4 Parallel Processing

Block-level Parallelization:

PARALLEL FOR each superblock:

 Place blocks in superblock

 (requires local grid copy)

Post-process:

 Merge results

 Resolve conflicts

Expected speedup: 2-3x on 4+ cores

6.5 Greedy Improvements

Best-First Search:

Instead of: Accept first valid position

Use: Search multiple positions, choose best

Search window: 20×20 around anchor

Positions evaluated: ~100

Selection: Highest score above threshold

Trade-off: 5x slower, 20-30% better placement

6.6 Constraint Relaxation

Progressive Relaxation:

threshold = -1

attempts = 0

WHILE NOT placed AND attempts < max_attempts:

```

    IF attempts % 20 == 0:
        threshold -= 1 // Relax threshold

    // Try placement with current threshold
    attempts++

```

Benefit: Higher placement rate (72% → 85%)

7. Implementation Details

7.1 Random Number Generation

Seeded RNG for Reproducibility:

```

import random

# Set seed for reproducible results
random.seed(42)

# Or use system time for true randomness
import time
random.seed(int(time.time()))

```

7.2 Floating Point Considerations

Distance Comparison:

```

# Avoid: direct float comparison
if distance == 10.0: # May fail due to precision

# Use: tolerance-based comparison
EPSILON = 1e-6
if abs(distance - 10.0) < EPSILON: # Robust

```

7.3 Grid Representation

Options:

Option A: 2D Array (Current)

```

grid = [[None] * width for _ in range(height)]
Access: grid[y][x]
Memory: O(W × H)
Access time: O(1)

```

Option B: Sparse Dictionary

```

grid = {} # (x, y) -> block_id
Access: grid.get((x, y))

```

Memory: $O(\text{occupied_cells})$
Access time: $O(1)$ average

Option C: Quadtree

For very large grids
Memory: $O(\text{occupied_cells})$
Range queries: $O(\log n + k)$

7.4 Block Sorting

Priority Order:

```
CATEGORY_PRIORITY = {  
    'ME': 0,          # Highest priority  
    'WE': 1,  
    'US': 2,  
    'SUPPORT': 3,  
    'SPECIALTY': 4   # Lowest priority  
}
```

```
blocks.sort(key=lambda b: CATEGORY_PRIORITY[b.category])
```

Rationale: ME blocks are most constrained, place first

7.5 Superblock Positioning

Anchor Calculation:

```
def get_superblock_anchor(superblock_index, grid_width, grid_height):  
    """  
    Distribute superblocks in 3-column layout  
    """  
    column = superblock_index % 3  
    row = superblock_index // 3  
  
    x = column * 30 + 5  
    y = row * 25 + 15  
  
    return (x, y)
```

7.6 Error Handling

Placement Failure:

```
try:  
    placed = place_block(block, x, y, grid)  
    if not placed:  
        log_warning(f"Could not place block {block.id}")
```

```

        unplaced_blocks.append(block)
except GridFullException:
    log_error("Grid capacity exceeded")
    break

```

7.7 Performance Profiling

Key Metrics to Track:

```

import time

start = time.time()
# ... algorithm execution ...
elapsed = time.time() - start

metrics = {
    'total_time': elapsed,
    'placement_time': placement_elapsed,
    'analysis_time': analysis_elapsed,
    'blocks_per_second': len(blocks) / elapsed,
    'placement_rate': len(placed) / len(blocks)
}

```

7.8 Validation Checks

Assertion-based Validation:

```

# After placement
assert len(placed_blocks) <= len(blocks)
assert all(b.placed for b in placed_blocks)
assert all(0 <= b.x < grid_width for b in placed_blocks)
assert all(0 <= b.y < grid_height for b in placed_blocks)

# Check no overlaps
for i, b1 in enumerate(placed_blocks):
    for b2 in placed_blocks[i+1:]:
        assert not overlaps(b1, b2)

```

8. Algorithm Variants

8.1 Variant A: Simulated Annealing

Concept: Accept worse placements with decreasing probability

```

PROCEDURE PlaceWithAnnealing(blocks, grid):
    temperature = T_initial
    current_state = RandomPlacement(blocks, grid)

```

```

best_state = current_state

WHILE temperature > T_min:
    new_state = Perturb(current_state)
    delta_score = Score(new_state) - Score(current_state)

    IF delta_score > 0:
        current_state = new_state
    ELSE:
        probability = exp(delta_score / temperature)
        IF random() < probability:
            current_state = new_state

    IF Score(current_state) > Score(best_state):
        best_state = current_state

    temperature *= cooling_rate

RETURN best_state
END PROCEDURE

Pros: Can escape local optima
Cons: Much slower (~100x)

```

8.2 Variant B: Genetic Algorithm

Concept: Evolve population of solutions

```

PROCEDURE PlaceWithGenetic(blocks, grid):
    population = GenerateInitialPopulation(size=50)

    FOR generation = 1 TO max_generations:
        scores = Evaluate(population)
        parents = Select(population, scores)
        offspring = Crossover(parents)
        offspring = Mutate(offspring)
        population = Replace(population, offspring)

    best_solution = max(population, key=Score)
    RETURN best_solution
END PROCEDURE

```

Pros: Very high quality solutions

Cons: Very slow (~1000x), complex implementation

8.3 Variant C: Constraint Programming

Concept: Use CP solver (e.g., Google OR-Tools)

```

PROCEDURE PlaceWithCP(blocks, grid):
    model = CreateCPModel()

    // Variables
    FOR each block:
        x[block] = IntVar(0, grid.width - block.width)
        y[block] = IntVar(0, grid.height - block.height)

    // Constraints
    FOR each pair (b1, b2):
        AddNoOverlapConstraint(x[b1], y[b1], x[b2], y[b2])

    FOR each rule:
        AddAdjacencyConstraint(rule)

    // Objective
    Maximize(SumAdjacencyScores)

    solution = Solve(model)
    RETURN solution
END PROCEDURE

```

Pros: Optimal or near-optimal solutions

Cons: Slow for large instances, requires external solver

9. Testing Strategies

9.1 Unit Tests

```

def test_can_place():
    grid = create_empty_grid(10, 10)
    block = Block(width=3, height=3)

    # Should succeed
    assert can_place(block, 0, 0, grid) == True
    assert can_place(block, 7, 7, grid) == True

    # Should fail (out of bounds)
    assert can_place(block, 8, 8, grid) == False

    # Should fail (occupied)
    mark_grid(grid, block, 0, 0)
    assert can_place(block, 0, 0, grid) == False

def test_euclidean_distance():

```

```

d = euclidean_distance((0, 0), (3, 4))
assert abs(d - 5.0) < 1e-6

def test_composition_match():
    target = {'me': 0.6, 'we': 0.4}
    actual = {'ME': 0.65, 'WE': 0.35}
    match = calculate_composition_match(target, actual)
    assert 80 <= match <= 95

```

9.2 Integration Tests

```

def test_full_workflow():
    blocks = generate_blocks(50)
    assert len(blocks) == 50

    superblocks = create_superblocks(blocks)
    assert len(superblocks) > 0
    assert sum(len(sb.blocks) for sb in superblocks) <= 50

    grid = initialize_grid(100, 80)
    placed, grid = place_blocks(blocks, superblocks, grid)

    assert len(placed) > 0
    assert len(placed) <= 50

    # Check no overlaps
    for b1 in placed:
        for b2 in placed:
            if b1 != b2:
                assert not overlaps(b1, b2)

```

9.3 Performance Tests

```

import pytest

@pytest.mark.benchmark
def test_placement_performance(benchmark):
    blocks = generate_blocks(100)
    superblocks = create_superblocks(blocks)
    grid = initialize_grid(100, 80)

    result = benchmark(place_blocks, blocks, superblocks, grid)

    # Should complete in reasonable time
    assert benchmark.stats['mean'] < 5.0 # 5 seconds

```

9.4 Property-Based Tests

```
from hypothesis import given, strategies as st

@given(st.integers(min_value=10, max_value=200))
def test_block_generation_count(count):
    blocks = generate_blocks(count)
    assert len(blocks) == count
    assert all(b.area > 0 for b in blocks)
    assert all(b.id.startswith('BLK-') for b in blocks)

@given(st.integers(min_value=2, max_value=10),
       st.integers(min_value=2, max_value=10))
def test_grid_dimensions(width, height):
    grid = initialize_grid(width, height)
    assert len(grid) == height
    assert len(grid[0]) == width
    assert all(cell is None for row in grid for cell in row)
```

10. Debugging Techniques

10.1 Visualization Debugging

Grid State Visualization:

```
def print_grid_ascii(grid):
    for row in grid:
        for cell in row:
            if cell is None:
                print('.', end='')
            elif 'BLK' in cell:
                print('B', end='')
            elif 'CORE' in cell:
                print('C', end='')
            else:
                print('X', end='')
        print()
```

10.2 Logging

Structured Logging:

```
import logging

logging.basicConfig(level=logging.DEBUG)
logger = logging.getLogger(__name__)
```

```

logger.debug(f"Attempting to place {block.id} at ({x}, {y})")
logger.info(f"Placed {len(placed)} / {len(blocks)} blocks")
logger.warning(f"Block {block.id} failed placement after {attempts} attempts")
logger.error(f"Grid overflow at superblock {sb.id}")

```

10.3 Assertion-Based Debugging

Invariant Checking:

```

def check_invariants(grid, blocks):
    # All placed blocks should have valid positions
    for block in blocks:
        if block.placed:
            assert block.x is not None
            assert block.y is not None
            assert 0 <= block.x < grid.width
            assert 0 <= block.y < grid.height

    # Grid cells should match block positions
    for y, row in enumerate(grid.cells):
        for x, cell in enumerate(row):
            if cell and cell.startswith('BLK'):
                block = find_block_by_id(blocks, cell)
                assert block is not None
                assert block.x <= x < block.x + block.width
                assert block.y <= y < block.y + block.height

```

Appendix A: Algorithm Comparison

Algorithm	Time	Space	Quality	Notes
Greedy (Current)	$O(n \times k \times R \times F)(W \times H + n)$		Good	Fast, simple
Simulated Annealing	$O(n \times T \times I)$	$O(W \times H + n)$	Excellent	Slow, tuning required
Genetic Algorithm	$O(P \times G \times n)$	$O(P \times n)$	Excellent	Very slow
Constraint Programming	Varies	$O(n^2)$	Optimal	External dependency
Random Placement	$O(n \times k)$	$O(W \times H + n)$	Poor	Baseline only

Legend: - n = blocks, k = attempts, R = rules, F = features - T = temperature iterations, I = inner loop - P = population size, G = generations

Appendix B: Parameter Tuning Guide

Parameter	Default	Range	Impact
max_attempts	100	50-500	Placement rate
threshold	-1	-10 to 5	Quality vs speed
grid_spacing	1	1-3	Density
superblock_size	20-30	15-50	Cohesion
distance_weights	See rules	-5 to 5	Adjacency strength

Tuning Process: 1. Baseline: Run with defaults 2. Identify bottleneck (time/quality) 3. Adjust relevant parameters 4. Re-evaluate metrics 5. Iterate

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Algorithm Verification: Required before production deployment

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